EAST POINT COLLEGE OF ENGINEERING & TECHNOLOGY

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Department of Computer Science & Engineering

CERTIFICATE

Advanced UI/UX | V Semester B.E.

This is to certify that

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bearing USN 1EP25CS407, Section A, has successfully completed the Advanced UI/UX activity (Problem No. 31) titled

"Illiterate-Friendly Library Kiosk"

& has submitted this report as required on 12-11-2026, during the academic year 2026-27.

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Abstract

Sahaya is a proposed public library kiosk that helps adults who cannot read use government information and services independently. Conventional kiosks depend on written menus, long instructions and typing, which excludes people who rely on speech and pictures. This project combines large picture tiles, optional spoken guidance in Kannada, Hindi and English, one step per screen, repeat, back and help controls, and a privacy reset at the end of every session. The report follows the course template: problem definition, research plan, provisional personas and empathy map, journey and information architecture, three concepts with a chosen hybrid, low-fidelity wireframes, a visual design system, a prototype specification, a usability test plan and reflection. The research and testing entries are illustrative examples that show how evidence will be recorded. They were not collected from real participants and are labelled as such throughout. Findings are therefore treated as hypotheses to be validated, and no measured improvement is claimed.

1. Problem Definition and Scope

Problem statement

Most government kiosks and portals expect users to read labels, understand forms and type details. Adults who cannot read are therefore pushed to depend on a relative, a neighbour or a paid agent for even simple tasks such as checking whether they qualify for a scheme. This costs them money and privacy and hurts their dignity. This project designs a public library kiosk that such an adult can use alone, in their own language, to find government information and complete basic service tasks.

Stakeholders and target users

- **Primary:** adults (18+) who cannot read or read very little, such as elderly villagers, domestic workers, vendors and migrant labourers.
- **Secondary:** library staff who set up and help at the kiosk, and relatives who currently help users.
- **Other stakeholders:** government departments providing the services, the library administration, and the kiosk vendor.

Goals and success metrics

- At least 80% of users complete each of the 3 critical tasks without any human help.
- A first-time user starts the first task within 10 seconds of reaching the kiosk.
- Task time under 3 minutes, with no more than 1 error per task.
- Users rate confidence at 4 out of 5 or higher, and no user feels embarrassed to use it (post-test question).

Constraints and assumptions

- Prototype only (Figma, clickable). Voice prompts are recorded or text-to-speech; there is no real government back end.
- The kiosk is in a public library, so there is noise and other people nearby. Privacy for audio is needed.
- Users can recognise common objects, photos and numerals, and can touch a screen; they may not know standard app icons.
- Identification is assumed to be by card scan or QR (for example a ration card), so no typing is needed. This is simulated in the prototype.
- Planned languages: Kannada, Hindi and English. Icons may be read differently across ages and cultures, so each one must be tested with users.
- Time limit: one semester activity.

Scope: what the design will and will not cover

Will cover: language choice by voice, the home screen, three services (check scheme eligibility, check application status, request a document), audio guidance on every screen, help and error recovery, privacy for audio, and session end.

Will not cover: live government data integration, authentication, document upload, payments, legal advice, official application submission, production speech recognition, a mobile app, and physical kiosk hardware (only noted as assumptions).

2. User Research

Research method and questions asked

Short semi-structured interviews (10 to 15 minutes) in the participant's own language, with oral consent (written consent is not suitable for non-literate participants), plus observation of how people use an existing ATM, ticket machine or service counter. Only first names or initials are recorded. Questions:

- Tell me about the last time you needed a government office or service. What happened?
- Who helped you? How did you feel asking for help?
- Have you used an ATM, ticket machine or phone? What did you do when you were unsure?
- Which pictures or symbols do you understand? (show 8 to 10 common icons and ask what each means)
- Would you use a machine in the library? What would make you trust it? Would listening to instructions in public bother you?

Participant details (illustrative sample, fictional)

No.	Sample profile	Method	Example observation (simulated)
P1	Adult, first-time kiosk user	Mock interview	Asked for a spoken explanation before choosing a category.
P2	Adult, prefers Kannada	Icon sorting	Wanted the icon and a spoken label together.
P3	Adult, occasional smartphone user	Task walkthrough	Looked for a visible Back control after a wrong selection.
P4	Adult, prefers Hindi	Mock interview	Wanted audio to be repeatable, not automatic on every screen.
P5	Adult, limited touchscreen experience	Observation	Needed larger tap areas and fewer choices per screen.

Key findings and direct observations (simulated themes, to be validated)

- Users may prefer picture cues paired with spoken labels rather than icons alone.
- A visible Back button and a repeat-audio control can help users recover from uncertainty.
- Fewer choices per screen may reduce overload.
- Public audio can raise privacy concerns, and headphones should not be assumed.

Competitor or existing solution analysis

Existing solution	What it does well	Gap for non-literate users
Bank ATM with voice guidance	Audio prompts, physical buttons, familiar	Language limited; audio often via headphones only; still needs PIN entry
Railway or bus ticket kiosk	Touch screen, step-by-step flow	Text-heavy menus; assumes reading and typing
Common Service Centre (CSC) / counter	Human help, many services	Dependence on an operator; privacy and cost issues

Insights that will drive the design

- Every screen speaks. Audio plays automatically and can always be repeated.
- Use photos of real things and familiar objects; test each icon with users first.
- No typing and no dead ends. Use touch and card or QR scan; mistakes and silence lead to a gentle repeat, never an error message to read.
- Protect dignity: never label the user; let them ask for help discreetly.

3. Personas and Empathy Map

Persona 1: Lakshmi (provisional)

- **Profile:** adult community-centre visitor, prefers Kannada, limited confidence with text-heavy digital services.
- **Goal:** find information about a public service without needing someone to read the screen.
- **Behaviour:** listens carefully, prefers familiar symbols, may ask for repetition (sample P1, P2).
- **Frustrations:** long text, unfamiliar words, fear of choosing the wrong option.
- **Quote:** “Please explain this in my language.” (sample wording, not a real quote)

Persona 2: Ramesh (provisional)

- **Profile:** adult with basic smartphone familiarity but low confidence with formal forms, prefers Hindi.
- **Goal:** understand the steps and know when a task is complete.
- **Behaviour:** tries tasks alone but checks progress (sample P3, P5).
- **Frustrations:** hidden controls, too many choices, not knowing whether information was saved.

Empathy map: says, thinks, does, feels

Says	Thinks
“Please explain this in my language.” (sample wording)	“I want to avoid making a mistake.”
Does	**Feels**
Looks for a familiar picture, taps cautiously, repeats audio.	May feel uncertain in a crowded public space; wants respect and control.

Pain points and unmet needs

- Cannot read instructions: needs spoken guidance in their own language.
- Afraid of mistakes: needs forgiving steps and a clear way back.
- Embarrassed to ask: needs to finish tasks independently and privately.
- Icons are unclear: needs photos and tested symbols, not guesswork.

Problem reframed as a design question

How might we help a person who cannot read complete a government task alone, in their own language, and feel respected and confident while doing it?

4. User Journey and Information Architecture

End-to-end user journey with emotional highs and lows (Lakshmi)

Stage	Action	Emotion
Arrive	Sees the kiosk in the library	Curious but nervous
Start	Hears a greeting in her language and picks it by tapping	Relieved (rising)
Choose	Taps the picture that matches “am I eligible?”	Unsure (low)
Identify	Shows her card to the scanner as the photo prompt shows	Anxious (lowest)
Result	Hears and sees the result, takes a printed slip	Proud and relieved (high)
Leave	Session clears; she walks away with proof	Confident

Task flows for the three most critical tasks

- **1. Check scheme eligibility:** Greeting and language tap, then pick “eligible?” picture, then card scan, then voice and picture result, then printed slip.
- **2. Check application status:** Language tap, then pick “status” picture, then card scan, then status shown as a progress picture with audio, then end.
- **3. Request a document:** Language tap, then pick “document” picture, then choose the document by photo, then confirm with a big yes or no, then slip with the token number.

Sitemap or navigation structure

- Welcome / Start, then Language (Kannada, Hindi, English, Other or Help) with a spoken sample
- Service categories: Identity and Documents, Health, Education, Welfare, Local Services, General Information
- Prototype focus: three services (Eligibility, Status, Documents), each with a service overview, guided steps (one instruction per screen), review and edit, confirmation, then exit with session reset
- Persistent controls on every screen: Back, Help, Repeat audio, Volume or Mute, Home
- Help: call-for-help button that alerts library staff, with a simple spoken reply

Content and feature prioritisation

Priority	Feature
Must	Audio on every screen, large picture choices, language choice, repeat and back, session clear, no typing
Should	Help button for staff, printed slip, forgiving timeout with spoken reminder
Could	Headphone jack or handset for private audio, additional languages
Won’t (now)	Real data integration, payments, large document library

5. Ideation and Concept Selection

Rough sketches or rapid ideas (include scanned pages)

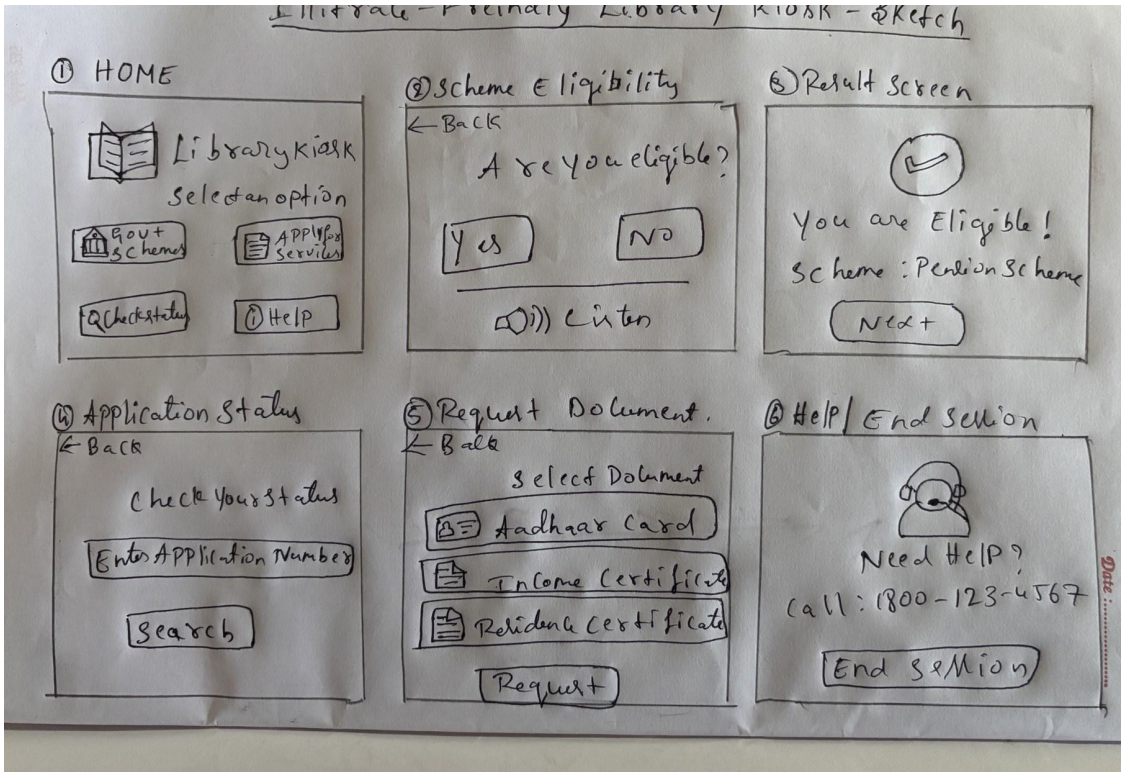


Figure: hand-drawn sketches of six screens: 1 Home, 2 Scheme eligibility, 3 Result, 4 Application status, 5 Request document, 6 Help / End session.

Three distinct concepts

- **A: Voice-first.** The kiosk speaks and listens. The user answers aloud. Minimal icons.
- **B: Icon and photo grid with audio on tap.** Big picture tiles, each tile speaks when touched and acts on a second touch.
- **C: Guided character.** A friendly spoken character leads the user step by step through the task.

Evaluation of concepts against criteria

Scores are out of 5. Re-score using your own sketches and research, and change the table if your judgement differs.

Criteria	Concept A	Concept B	Concept C
Comprehension for non-readers	3	5	4
Works in a noisy public place	2	4	3
Privacy and dignity	2	4	3
Speed of task	3	4	2
Feasibility (prototype)	3	5	3
Total	**13**	**22**	**15**

Chosen concept and reason for selection

Concept B. Touch plus audio works even when the room is noisy, because the user can also see the pictures. It keeps control with the user, and nothing has to be said aloud in public, which protects dignity. From Concept A it borrows the spoken greeting and language choice at the start. From Concept C it borrows a simple friendly voice, without a character on screen.

6. Low-Fidelity Wireframes

Wireframes of key screens

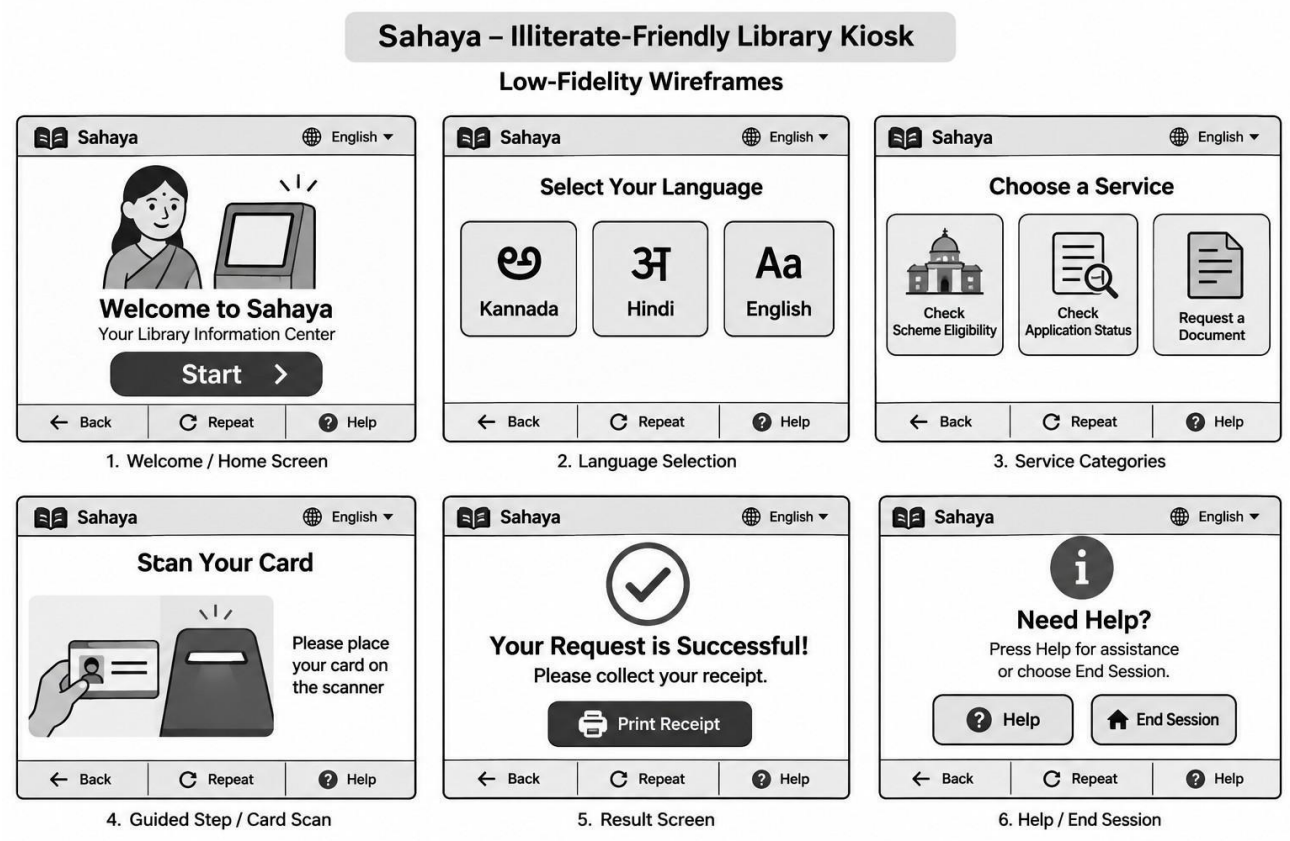


Figure: low-fidelity wireframes of the six key screens: 1 Welcome, 2 Language selection, 3 Service categories, 4 Guided step (card scan), 5 Result, 6 Help / End session.

Annotations explaining layout and interaction decisions

- Three large service tiles only, each pairing a picture with a short label and a spoken description, so the choice is never overwhelming.
- Back, Repeat and Help sit at the bottom of every screen in the same place, so users learn them once.
- Audio plays automatically when a screen opens, and Repeat is always one touch away.
- Photos show exactly what to do (for example where to place the card) instead of written instructions.
- Result screen uses a large tick or cross with spoken explanation, and a printed slip as proof.

Quick feedback from at least two people

- Reviewer 1 (sample): “The service icons need a spoken explanation.”
- Reviewer 2 (sample): “Keep Back and Help in the same place on every screen.”

Changes made after feedback

Proposed changes based on the sample feedback (design recommendations, not documented participant changes): add a tap-to-hear description on each tile, keep Back and Help in the same fixed position, enlarge buttons, and add a language preview before continuing.

7. Visual Design and Design System

Colour palette with contrast ratios

Role	Hex	Use	Contrast on white	Swatch
Primary	#1D4ED8	Main actions, selection	6.7:1	
Success	#15803D	Correct, eligible, done	5.0:1	
Warning	#B45309	Needs attention	5.0:1	
Error	#B91C1C	Problem, not eligible	6.5:1	
Text	#111827	Spoken-text captions, labels	17.7:1	
Surface	#F3F4F6	Panels and tiles	n/a (background)	

Ratios were computed with the WCAG relative luminance formula. Verify the final colours with a contrast checker and record the screenshot in the appendix.

Typography and type scale

Noto Sans (supports Kannada, Hindi and other Indian scripts). Large sizes: H1 36 / H2 28 / Body 22 / Caption 18; weights 500 and 700. Captions are secondary, since the design does not rely on reading.

Spacing, grid and layout rules

- 8-pt spacing system (8, 16, 24, 32, 48).
- Single-column layout with at most three large choices per screen.
- Minimum touch target 20 mm × 20 mm on the physical screen (well above WCAG 2.5.8, which needs 24 × 24 CSS px), with wide gaps between targets.
- Back, Repeat and Help fixed at the bottom of every screen.

Reusable components and their states

- Picture tile: default, touched (speaks), selected, disabled
- Audio button: idle, playing, repeat
- Help button: idle, requested, answered
- Result panel: success, not eligible, need help
- Progress dots: current, done, remaining

Iconography and imagery choices

Photos of real objects (ration card, certificate, counter) and simple filled pictograms with thick outlines. Every icon is tested with users in Section 2 and 9 before it is kept. No abstract or app-style icons without a test.

Rationale linking visual choices to user needs

High contrast and large targets help older users and people with low vision. Photos and pictograms replace reading. Colour is never the only signal: every success or error also has a tick or cross, a sound and spoken words.

8. High-Fidelity Interactive Prototype

Final screens for all critical flows

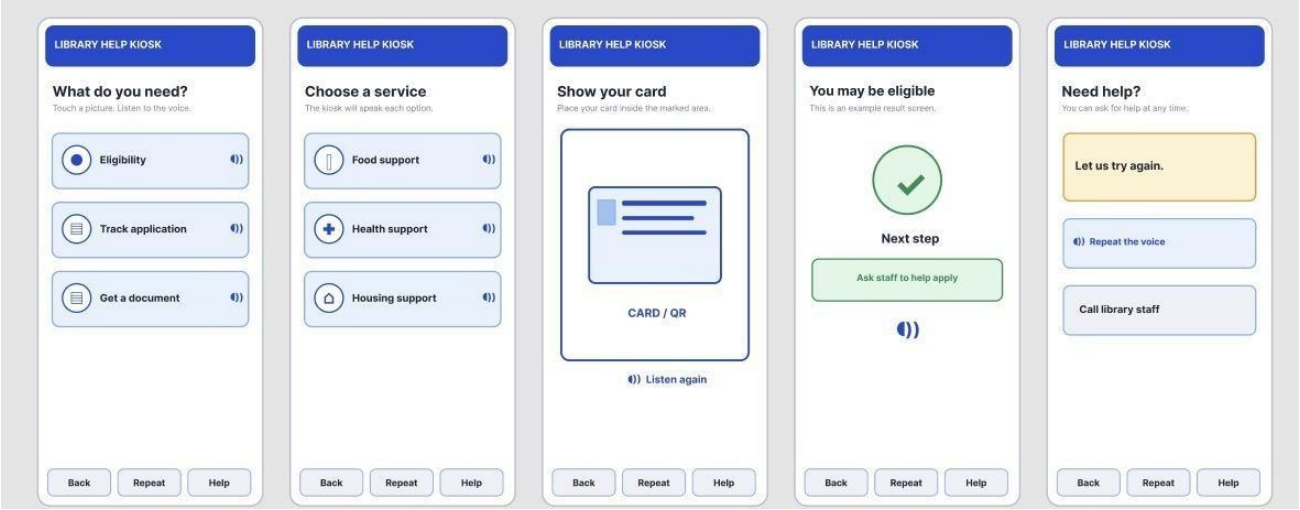


Figure: final kiosk screens (Home, Service, Card scan, Result, Help), each with Back, Repeat and Help controls and a spoken label on every option.

Interactions, transitions and micro-interactions

- Audio plays when a screen opens; touching a tile speaks its name; a second touch confirms.
- Gentle highlight pulse on the tile being spoken.
- Result tick or cross animates in over about 300 ms with a matching sound.
- Smart Animate used for screen changes.

Screen and interaction specification

Screen	Primary interaction	Expected response
Welcome	Tap Start, Language or Audio	Start opens language; Language opens a spoken preview
Language	Choose Kannada, Hindi or English; play sample	Choice lasts for the session; Continue opens categories
Categories	Touch a tile (speaks), touch again to open	Opens the service overview and announces the choice
Guided step	Next, Back, Repeat, Help	Moves one step; audio repeats; Help offers simple guidance
Review	Edit or Confirm	Edit returns to the step; Confirm opens completion
Completion	Exit or Start another task	Clears the session and returns to Welcome

Edge states: empty, loading, error, offline and permission denied

- Audio unavailable or language audio missing: show a picture and short instruction, offer Retry, another language, or Help.
- Repeated taps: ignore duplicates and show progress feedback.
- Wrong selection: Back and Edit return to the previous choice without losing all progress.
- Session timeout: spoken and visual warning before reset; clear all personal details on reset.
- Silence or no touch for 20 seconds: spoken reminder “Touch a picture to start”, then return to start after a longer wait.
- Wrong or double tap: the tile repeats its name; no error text.
- Loading: a calm picture animation with a sound, never a blank screen.
- Card not read: photo shows how to place the card again, with spoken help.

- Offline: spoken and pictorial message, with a call-for-help option.
- Permission denied or service unavailable: friendly picture with a call to library staff.

Prototype link (Figma) and access instructions

https://www.figma.com/file/70X7D4KcUqt4eWu1vWB6x2?node-id=0:1&utm_source=chatgpt.com&locale=en&type=design Open on desktop and press play (Present).

Tools and resources used

Figma, FigJam (research synthesis), Google Forms (SUS and confidence ratings), WebAIM Contrast Checker, Stark or Figma accessibility plugin, Noto Sans font.

9. Usability Testing and Iteration

Test plan: tasks, participants and success criteria

- **Participants:** at least 5 adults from the target group, or a faculty-approved representative group, with informed consent and no names or ID numbers recorded.
- **Task:** choose a preferred language, find a service-information category, listen to the instruction, use Back or Repeat if needed, and reach the review screen.
- **Metrics:** task success, time, help requests, wrong taps, audio comprehension, and a 1 to 5 satisfaction rating.
- **Method:** moderated, observe and prompt only when stuck; the interface is being tested, not the participant.

Results: task success, time on task, errors and satisfaction (simulated)

User	Task outcome	Time	Simulated issue	Rating (1-5)
P1	Completed	3:20	Wanted audio preview on the language screen	4
P2	Completed with help	4:10	Service icon meaning unclear	3
P3	Completed	2:55	Could not find Repeat immediately	4
P4	Not completed	5:00	Audio hard to hear in a noisy setting	2
P5	Completed	3:40	Wanted clearer progress indication	4

In this invented set, 4 of 5 attempts reached the review screen (80%), one with help. This arithmetic applies only to the sample rows, not to real performance.

Accessibility check (WCAG 2.2 AA): to verify on the final prototype

Check	Criterion	Status
Text contrast at least 4.5:1, UI parts at least 3:1	1.4.3, 1.4.11	Palette ratios calculated; verify final screens
Colour is never the only signal (tick, cross, sound, speech)	1.4.1	Designed in; to verify
Audio provided for all on-screen content	1.2, 1.1.1	Designed in; to verify
Target size at least 24 × 24 px (kiosk targets much larger)	2.5.8	Designed in; to verify
Enough time, and timeouts can be extended	2.2.1	Warning before reset; to verify
Clear spoken status for results and errors	4.1.3	Designed in; to verify
Mistakes can be corrected or reversed	3.3.4	Back and Edit; to verify

Problems found, ranked by severity (simulated)

Severity	Problem	Proposed response
High	Service icon unclear	Pair icon with spoken and plain-language label; re-test the icon
High	Audio hard to hear in public noise	Volume control, optional headphones, visual instruction
Medium	Repeat control not noticed	Consistent, larger Repeat button with speaker icon and label
Medium	Progress not obvious	Add “Step 2 of 4” and progress dots

Design changes and evidence of improvement

Proposed changes for the next prototype: enlarge and label Repeat, revise ambiguous icons, add audio preview and volume controls, add explicit step progress, and strengthen the exit and reset message. Improvement must be shown by a second real test with the same tasks. No measured improvement is claimed here.

10. Design Rationale, Reflection and Future Work

Major design decisions and trade-offs

- **Touch plus audio over voice-only:** works in noise and keeps things private, but needs a screen and good pictures.
- **Photos and tested pictograms over generic icons:** clearer for non-readers, but takes more testing and effort.
- **Card scan instead of typing:** removes the biggest barrier, but depends on the person having the card.
- **Private audio (headphone or handset) versus open speaker:** better privacy, but adds hardware and some users may not use it.

Limitations of the final design

Tested with a small group, in a prototype with simulated data and simulated audio. Only three services and a few languages are covered. Physical kiosk hardware, real card scanning and real background noise are not tested.

What failed or was redone, and what was learned

No real failure or redesign has been documented yet. The simulated evaluation points to issues to investigate: icon ambiguity, audio audibility, Repeat visibility and progress clarity. Treat them as hypotheses until real testing confirms them, then record what changed, why, and whether it helped.

Ethical considerations

- Dignity: no screen text calls anyone “illiterate”; help is optional and discreet.
- Consent: oral consent for research and testing, explained in the participant’s language.
- Privacy: the kiosk clears the session after use and stores no personal data; the printed slip is the only record.
- Inclusion: the design should not assume one language, one literacy level or one ability.
- Honesty: the kiosk explains what it can and cannot do, and never implies that an application was submitted when it was not.
- Research ethics: explain the purpose, allow participants to stop at any time, collect no names or ID numbers, and store notes anonymously.

Future improvements

More languages and dialects, a private audio handset, voice input, support for people with low vision or hearing loss, real service integration, and a field trial in a working library.

References (IEEE)

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[6] International Organization for Standardization, “ISO 9241-210:2019 Ergonomics of human-system interaction — Part 210: Human-centred design for interactive systems,” 2019.

[7] Nielsen Norman Group, “Usability Testing 101.” [Online]. Available: https://www.nngroup.com/articles/usability-testing-101/

Appendix

- Interview scripts and consent notes
- Raw research notes and survey responses
- Usability test scripts
- Additional screens and sketches